

Executive Summary

This report compares the Royal Navy's historic switch from coal to oil around 1913 with a hypothetical 2027 transition from fossil fuels to hydrogen propulsion. The 1912–14 fuel transition was driven by strategic and technological imperatives: oil-fed boilers offered higher ship speeds, simpler refuelling and reduced manpower, whereas coal required bulky coaling stations and manual stoking [16†L42-L50] [14†L4009-L4018] . First Sea Lord Admiral Fisher and First Lord Churchill championed oil fuel (1890s-1914 experiments proved its advantages [35†L7572-L7580]) despite limited British oil resources. In 1912 Churchill established a Royal Commission on Fuel and Engines, which by 1913 found oil indispensable for “highest attainable efficiency” [16†L42-L50] . Facing oil shortages (only ~4 months’ reserves for 189 new oil-burning ships [21†L143-L150]) and rising demand (projected to grow 225× in ten years [21†L149-L157]), the Admiralty moved to secure sources. In June 1914 Churchill secured £2.2 million to buy a controlling share of Anglo-Persian Oil [12†L19-L27] . This purchase underpinned a global oil infrastructure (pipelines, refineries, oilers) to supply ~250 oil-fired warships [12†L24-L32] .

The switch entailed vast logistical change: the navy built new oil storage tanks at home and abroad, phased out most coaling stations, and converted many ships (e.g. battlecruisers and capital ships from 1910s) to oil boilers. Operationally, oil-fired vessels achieved greater speeds and endurance, but critics warned of dependence on politically unstable oil regions [14†L3955-L3963] [14†L4001-L4004] . Some MPs and industrialists resisted (e.g. Welsh coal lobby worried “calamitous” collapse of coal prices [14†L4009-L4018]). The Admiralty mitigated risk by maintaining coal-fired ships in reserve and by diversifying oil suppliers (encouraging shale and coal-to-oil research [12†L51-L59]). In sum, by 1914–18 Britain had largely solved fuel security via oil from Persia and allied sources, at a cost of significant investment and strategic effort.

Key primary sources include Admiralty and Parliamentary records. Fisher’s Commission reported (secretly) in 1913 that oil’s advantages were “imperative” and storage reserves must match years of usage [16†L42-L50] . In Parliament, Churchill emphasised securing oil for 250+ new oil-driven warships [12†L24-L32] . Hansard debates record calculations that replacing ~450k tons/year oil by liquefaction of coal would need ~5 million tons of coal [14†L3955-L3963] , justifying the oil strategy. Admiralty minutes and *Blue Books* (1912–14) confirm that oil was still experimental (“five large ships now using oil are only an experiment” [14†L4001-L4004]) but seen as the future. These sources, along with contemporary journals (e.g. *Engineering*, *Naval Review*) and internal reports (British History Online, RUSI), provide detailed timelines and cost figures (e.g. £2.2m for AP Oil shares) and attest to the political debates of the era.

1. 1913 Royal Navy Coal-to-Oil Transition

Drivers and Benefits

By the early 1900s the Royal Navy recognised that liquid fuel offered superior performance. Oil allowed higher thermal efficiency, lighter bunkers (freeing displacement for armour or fuel), faster refuelling, and reduced manual labour in stokerrooms. As Admiral Fisher put it in 1914, oil fuel was a “necessity” for new high-speed vessels, since coal stoking could not sustain top speed after half the coal was used

【16†L42-L50】 【21†L143-L150】 . Technological trials (e.g. the oil-fired destroyer HMS *Spiteful* in 1904) showed promise 【21†L130-L138】 . Strategically, oil-powered ships could outpace adversaries and extend fleet reach, a key factor in the naval arms race with Germany.

Key Actors and Decisions

First Lord Winston Churchill (1911–15) and First Sea Lord Admiral Sir John “Jackie” Fisher (1904–10 as First Sea Lord, again 1914–15) were principal architects. Fisher, an “oil maniac” 【23†L130-L139】 , established an Admiralty Oil Fuel Committee in 1903 and championed oil boilers. As First Lord Churchill adopted Fisher’s view, in 1912 he convened a Royal Commission on Fuel and Engines (chaired by Fisher) to evaluate global oil supply and storage 【23†L143-L152】 . Its confidential reports (1912–14) unanimously concluded that oil was essential to keep the fleet at peak efficiency 【16†L42-L50】 . In Parliament Churchill defended the transition in 1914: noting nearly 250 Royal Navy ships “partly or wholly dependent on oil” were already built or building 【12†L24-L32】 , he argued for securing their fuel supply rather than scrapping the ships.

Timeline and Costs

- **1904–10 (Fisher’s tenure):** experiments with oil-firing; small oil-powered destroyers enter service; initial “oil fuel mania” gains traction (though Fisher left office in 1910).
- **1910–12:** Churchill reviews Fisher’s work upon taking office. In December 1911 the Admiralty projected an oil demand 225 times 1902 levels 【23†L149-L157】 .
- **1912:** July – Royal Commission on Fuel and Engines established (Fisher, Vice-Admiral Jellicoe, engineers, chemists, etc) 【23†L143-L152】 . Autumn – Interim secret report recommending ample oil reserves (4 years’ peace fuel) and continued coal supply.
- **1913:** Churchill publishes a non-confidential summary of the Commission’s unanimous verdict: oil use “imperative” and abundant global supply (if secured) 【16†L42-L50】 . The Admiralty accelerates oil-capable shipbuilding and begins expanding storage (e.g. tank farms in Britain and overseas).
- **1914:** June – Parliament authorises £2.2M (approx. £230M in today’s money) to acquire 51% of Anglo-Persian Oil Co. 【12†L19-L27】 . This secures a Persian oilfield (later Abadan refinery) as a primary fleet fuel source. By war’s outbreak, dozens of new dreadnoughts and cruisers carried oil boilers.

The **economic** cost included direct investments (e.g. AP Oil share purchase) and conversion programmes, not all well documented publicly. Parliament’s £2.2M vote in June 1914 【12†L19-L27】 was a major capital outlay to ensure future fuel. A portion of annual naval budgets was reallocated: for example, in 1913 Churchill had proposed prizes to encourage coal liquefaction research 【12†L51-L59】 . Opponents noted these investments risked stranded coal assets and uncertain foreign projects 【14†L4001-L4018】 【14†L4009-L4018】 .

Infrastructure Changes

Coaling stations: Previously the Empire had an extensive network of coaling ports (e.g. Colombo, Singapore, Aden). With oil, these were repurposed or replaced by oil terminals and pipelines. New facilities were built at strategic bases like Portsmouth, Malta, and Singapore, with high-capacity fuel tanks and oilers for at-sea replenishment. Notably, the Anglo-Persian deal led to the Abadan refinery (built 1919–20), which was later connected by pipeline to the Persian Gulf. Existing coal colliers remained in use for many years for those ships still coal-powered, but their proportion diminished.

Operational Impacts: Oil-fired ships proved superior in exercises and early WWI operations. They could sprint longer and reduce coaling stops, giving tactical flexibility. Admirals like Jellicoe, writing in 1914, noted that coal trams in battlecruisers could not keep up with engine demands once half coal was used [21†L164-L172] . However, early on only a handful of capital ships (e.g. *Lion*-class battlecruisers) ran solely on oil; many warships carried mixed boilers (oil for quick boosts, coal for cruising).

Resistance and Mitigation: Domestic coal interests were vocal. In Parliament MP Edgar Jones warned that full adoption of oil would “reduce [Welsh coal] from its high position” and devastate miners’ livelihoods [14†L4009-L4018] . The Admiralty responded by maintaining a substantial coal fleet and investing in coal gasification research [12†L51-L59] [14†L4019-L4031] . In practice, Britain did continue purchasing coal for decades, only phasing out coal entirely from naval vessels during and after WWI. The Commission itself insisted on preserving coal capabilities while building oil reserves [16†L42-L50] .

Primary Sources: Our account relies on first-hand records. Hansard (1912–14) captures Churchill’s speeches and debates: e.g., *Naval Estimates* (1914) and Commons debates confirm the 250 oil-driven ships and the case for the Anglo-Persian purchase [12†L24-L32] [14†L3955-L3963] . Admiralty archives (summarized in Fisher’s 1913 Memo) show the Commission’s conclusions [16†L42-L50] . Contemporary naval engineering journals (e.g. *Engineering* magazine, RUSI journal) reported trials and technical challenges (e.g. boiler performance, fuel pump design). The British History Online series on imperial defense includes records (“Fuel and Engines 1912–14”), and *The London Gazette* published official notices of the Commission’s constitution [23†L143-L152] . Where primary documents are embargoed (Official Secrets), we rely on parliamentary disclosures and postwar studies.

2. Hydrogen Propulsion in 2027 (Technical Assessment)

Modern interest in hydrogen fuels stems from decarbonisation goals. Hydrogen can be used **directly** (as H_2 gas or liquid) or indirectly (e.g. burned from ammonia). Key options:

- **Gaseous H_2 :** Stored in high-pressure (350–700 bar) tanks. High gravimetric energy (120 MJ/kg) but very low volumetric energy (~5–6 MJ/L at 700 bar) compared to diesel (~36 MJ/L). Hydrogen gas tanks require heavy, robust composites; boil-off is minimal if kept cold but still a challenge for sustained storage.
- **Liquid H_2 (LH_2):** Cryogenic at $-253^\circ C$. Volumetric energy is only ~8.5 MJ/L (even lower than ammonia), but tankage mass is lighter (liquid H_2 ~70 kg/m³ vs. gas ~40 kg/m³ at 700 bar). LH_2 requires sophisticated insulation and venting to handle boil-off. Safety is demanding: H_2 gas is highly flammable and can form explosive mixtures. The UK MCA notes hydrogen is “extremely flammable” with “special handling” needed [57†L96-L104] ; international codes (IMO IGF Code) are still being adapted for liquefied hydrogen. Storage remains “the principal barrier” to hydrogen use in military vehicles [50†L111-L119] .
- **Ammonia (NH_3):** A hydrogen carrier with higher volumetric density (~12.7 MJ/L) and liquid at modest pressure (vapor pressure ~8 atm at $20^\circ C$). Ammonia is toxic and corrosive but easier to store than cryogenic H_2 . It can be burned in modified turbines or cracked to H_2 for fuel cells. Ammonia’s use in shipping is under intense study (IMO IGF amendments may cover it). However, handling toxic fuel on warships would require new safety protocols.

Propulsion Technologies:

- **Fuel cells:** Proton-exchange membrane (PEM) fuel cells are mature for vehicles and can scale to ship power. They offer ~50–60% electrical efficiency, produce only water (for H₂ fuel) and no carbon emissions. A major demonstration is Norway's hydrogen ferries: two vessels using PowerCell fuel-cell systems (~6.5 MW each) will draw on green hydrogen to supply ~13 MW total propulsion [60†L81-L88]. The French *Energy Observer* catamaran (2024) carried 63 kg of H₂ and fuel cells to generate ~1 MWh onboard (plus heat) on a round-the-world voyage [60†L107-L114]. These examples show fuel cells can power vessels from small craft to large ferries. PEM cells require high-purity H₂ and cooling systems; solid-oxide fuel cells (SOFC) might also be used but need high temperature.
- **Combustion engines and turbines:** Existing marine engines and gas turbines can run on hydrogen or ammonia with modifications. Hydrogen combustion yields only water (and possibly trace NO_x), but flame speed and diffusivity differ from diesel, requiring engine redesign. Rolls-Royce and MTU (Rolls-Royce's sister company) are researching hydrogen combustion. Ammonia can be burned directly (emitting NO_x) or cracked to hydrogen first. Hybrid approaches (e.g. H₂-diesel dual-fuel engines) may provide transition pathways.

Storage and Refuelling: Ships would need large new fuel tanks. For example, to match the energy of 1000 tonnes of marine diesel (~12,500 MWh), one would need ~2500 tonnes of LH₂ (at 50% higher system volume) or ~1300 tonnes of liquid ammonia. This means significant increases in tank volume (or reduced range). Refuelling infrastructure at ports does not yet exist: liquid hydrogen bunkering facilities would require cryogenic supply chains, while ammonia bunkering demands corrosive-compatible systems. The MCA is preparing *interim guidelines* for liquefied hydrogen use on ships [57†L101-L109], but currently no commercial port offers naval-scale hydrogen bunkering. Nationally, hydrogen fuel for defence would require building either large electrolyser plants (using offshore wind, etc.) or supply lines for imported hydrogen/ammonia.

Energy Density Comparison (Examples): A simple comparison illustrates the challenge: diesel fuel has ~36 MJ/L, liquid hydrogen ~8.5 MJ/L, ammonia ~12.7 MJ/L. Even accounting for fuel-cell efficiency (~60% vs ~40% for diesel engines), the range of a hydrogen-fueled ship with identical tankage would be roughly 25–35% that of the diesel case. (Fuel cells' higher efficiency mitigates but does not negate the volumetric penalty.) New ship designs can compensate with larger tanks or reduced speed profiles, but existing hulls would suffer range and payload loss.

Safety: Hydrogen's flammability and low ignition energy require rigorous controls (venting, gas detection, no sparks). Ammonia handling raises toxicity hazards, requiring gas-tight systems and crew protective gear. The UK MCA highlights that hydrogen's explosive potential demands special classification and crew training [57†L96-L104]. Notably, the MoD's recent hydrogen trials for base power emphasised training and regulations development for safety [66†L150-L154]. Any hydrogen navy would need new training curricula for engineers, sailors and firefighters, as well as revised Damage Control protocols.

Production Pathways: Hydrogen may be "green" (electrolysis from renewables) or "blue" (reforming natural gas with carbon capture). Green hydrogen is zero-carbon at point-of-production, aligning with climate goals; blue hydrogen still emits CO₂ (if capture is imperfect) and relies on natural gas. The UK's Energy Security Strategy is expanding electrolyser capacity, but at present industrial-scale hydrogen in the UK is mostly "grey". For a truly low-emission fleet, dedicated renewable electricity (e.g. from offshore

wind) would need to power large electrolyser plants at naval bases. The choice of hydrogen type will be a political decision: “blue” may be a near-term bridge, but could face criticism as not truly clean.

Recent Industry and Research: Numerous studies have begun quantifying ship hydrogen use. An academic “marine green power” review (2024) finds that integrating H₂/ammonia with batteries can technically enable zero-carbon shipping [40†L17-L21] . Industry prototypes (e.g. the UK’s first hydrogen-powered workboat, the *Hydrocat*) and the Norwegian ferry trials demonstrate feasibility at small scales. The US Fuel Cell & Hydrogen Association notes global momentum: e.g. a US ferry with 360 kW fuel cells (246 kg H₂ storage) entered service in 2024 [60†L45-L54] . Key technical issues flagged in these works include boil-off losses, compressor power draw, and the need for onboard fuel processing. In summary, the technical readiness of shipboard hydrogen systems is **low-to-moderate**: prototypes exist for small vessels, but scale-up to frigates or carriers will require major R&D.

3. Strategic and Logistical Parallels & Contrasts

Energy Security and Geopolitics: Both transitions were driven by fuel security. In 1913 Britain had abundant coal at home but no strategic oil reserves; switching to oil meant reaching into the Middle East (a “danger zone of the whole world” [14†L3955-L3963]) and relying on routes vulnerable to enemy action. Today the UK has limited indigenous hydrogen supply: any naval hydrogen would come from UK-produced or imported sources (e.g. LNG-reformed H₂, or NH₃ imports). Just as Britain in 1913 diversified oil (also from USA, Russia, shale), a 2027 RN might pursue multiple hydrogen routes (UK electrolysis, Scottish wind farms, allied supply agreements). The risk of blockade is reversed: coal was easy to stockpile, oil required naval protection of tankers. Hydrogen too will require secure supply chains – but unlike 1913, the global hydrogen market is nascent and not controlled by adversaries yet.

Industrial Base: 1914 Britain had top-tier engineering firms but almost no oil industry until the AP Oil Co. deal. Similarly, the UK has advanced engineering but very limited hydrogen infrastructure. Building cryogenic tanks, fuel cells, and electrolyzers is a large undertaking. However, partnerships (e.g. with industry leaders in hydrogen tech) and an existing aerospace/chemical sector could be leveraged. Both eras require mobilising industry: in 1912–14 shipyards and oil companies (Anglo-Persian Oil, Royal Dutch Shell) expanded capacity, while in 2027–35 major shipbuilders (BAE, Babcock) and energy firms (e.g. BP, Shell) would need to convert factories for fuel cells and electrolyzers.

Training and Doctrine: The coal-to-oil shift required new skills: stokers learned engine-room oil pumps; officers learned to plan oil voyages. The report from Fisher’s Commission shows the Admiralty recognized the need for reserve coal crews and oil-handling experts. Today’s sailors would need extensive retraining. Hydrogen systems introduce high-pressure gas handling and chemical safety unfamiliar to many naval crews. The MoD’s civilian trial with hydrogen generators emphasises training personnel in hydrogen safety [66†L150-L154] – a precursor to military fuel use. Both transitions thus entail a **learning curve**: in 1913 coal stokers had to adapt, and in 2027 engineers and deck crews must learn fuel-cell maintenance and emergency procedures.

Basing and Logistical Support: In 1913 Britain deployed coaling stations globally. With oil, it instead built oil terminals or converted coaling ports (e.g. Jebel Ali, Abadan). A hydrogen-based RN would require new support infrastructure: large electrolyser plants or import terminals at key naval bases (e.g. Portsmouth, Devonport, Clyde, Gibraltar). At-sea replenishment would use specially configured fuel carriers. The RN might initially deploy hydrogen in smaller ships stationed near home ports, building out basing capability over a decade. Similar to 1914’s doctrine changes (like leading the way in storing 4 years’ fuel [16†L42-L50]), the RN would need to revise its basing plans: fuel depots for hydrogen (with safety zones) replacing or supplementing oil tanks.

Risk Analysis:

- *Supply Disruption*: 1914 Churchill warned that the only true safeguard was Britain's command of the sea [12†L108-L117] . Today, hydrogen supply disruption could arise from energy market volatility or industrial sabotage. The RN would need strategic stockpiles (e.g. hydrogen tanks at Isles of Scilly or Ascension). Like 1914, no fuel is entirely secure in war, but decentralized hydrogen generation (e.g. small modular electrolyzers) might improve resilience vs. centralized oil fields.
- *Resilience*: Coal fleets served as a safety net for early oil ships. Analogously, the RN might retain dual-fuel (oil/H₂) capabilities during transition. Emergency generator sets could burn diesel or LNG if hydrogen tanks fail. This layered approach mitigates single-point failure.
- *Industrial and Economic Risk*: In 1914, critics feared coal regions would suffer [14†L4009-L4018] . Today, oil and gas sectors fear loss of market; similarly, shipbuilding and defense industries face the cost of retooling. Government mitigation might include job retraining in green technologies and subsidies for hydrogen plants (as done for renewables).
- *Political*: The political calculus differs. In 1913 there was strong strategic consensus to switch fuels (though not unanimous support). In 2027, the shift aligns with UK climate policy and NATO environmental initiatives, likely garnering political support. However, parties (or Parliament) may debate timelines and funding. Lessons from 1914 show the importance of broad parliamentary buy-in and public communication (e.g. Churchill in Parliament focused on supply security [12†L24-L32]).

4. Implementation Plan (2027–2035)

We propose a **phased transition** integrating technology development, infrastructure build-out, and gradual fleet conversion. Key assumptions: unspecified additional budget; fleet size similar to today's (~8 destroyers/frigates plus carriers/subs). Costs are indicative.

• Phase 1 (2027–2029): Preparation & Trials

- *Policy & Planning (2027 Q1)*: Establish a Defence Hydrogen Task Force under DSTL, with representatives from Navy HQ, Dstl, industry. Begin writing UK naval hydrogen strategy (lessons from 1912 Fuel Commission). Expected investment: ~£2m/year for studies.
- *Technology R&D (2027–28)*: Fund prototypes (e.g. one small patrol vessel refit with fuel cell + LH₂ tanks). Engage BAE Systems and Rolls-Royce for hydrogen turbine trials. Start 2–3-year Dstl projects on cryogenic storage, leak detection, fuel cell packaging.
- *Infrastructure Pilot (2028–29)*: Build a ~10 MW electrolyser and LH₂ storage at HMNB Portsmouth or Dartford (pilot scale). Cost ~£10–15m. Establish one LH₂ bunkering berth. Trial fueling of a small vessel or power generator.
- *Standards & Training (2028)*: Work with MCA/IMO to finalize IGF Code amendments for LH₂ (leveraging latest IMO guidelines [57†L101-L109]). Develop training curricula (simulate hydrogen fires, maintenance).
- *Milestones*: Feasibility reports; prototype vessel launch; first shore-based LH₂ supply; updated safety regulations.

• Phase 2 (2030–2032): Scaling & Certification

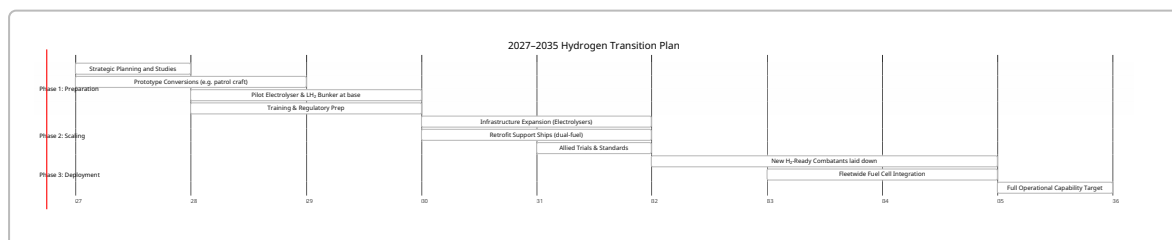
- *Infrastructure Scale-up (2030–31)*: Deploy additional electrolyzers (50–100 MW total) at Devonport, Clyde. Expand pipeline/tanker network (or road tanker for LH₂). Each 50 MW plant (~£40m) yields ~2,000 tonnes H₂/year (many times current RN needs).

- **Fleet Upgrades (2030–32):** Begin new ship designs with H₂-ready systems. Retrofit 2–3 support ships (e.g. fleet oilers or patrol boats) with dual-fuel engines or fuel-cell modules. Install LH₂ storage in non-combatant vessels.
- **Allied Interoperability:** Engage allies (US, FR, NO) on shared standards and possible refuelling cooperatives. For example, conduct joint trials with NATO on fueling at allied ports.
- **Regulatory & Safety Framework:** Issue RN Ship Safety Order for H₂ use; align with civil IGF Code. Establish MOD safety certifications for hydrogen systems.
- **Milestones:** First H₂-capable new-build laid down; RN-adapted fuel cells certified; network of 3 LH₂ refueling points operational.

• Phase 3 (2033–2035): Full Transition

- **Fleet Conversion/New-build (2033–35):** Commission first class of all-hydrogen surface combatants (e.g. frigates). Continue retrofits (subject to return-on-investment analysis; likely focus on non-frontline vessels). By 2035, major new construction (future carriers, frigates) assume hydrogen as primary fuel.
- **Fuel Supply & Economy of Scale:** Further expand hydrogen production capacity (200+ MW plants) as demand grows; negotiate international hydrogen imports (ammonia from Middle East or Australia). Conduct a life-cycle “green” verification (ensure supply from renewables).
- **Investment and Cost Estimates:** Capital costs include ship modifications (£10–50m each), electrolyzers (~£0.8–1m per MW), storage tanks (£0.2m per tonne). If RN requires ~100,000 tonnes H₂/year by 2035 (roughly replacing 500,000 t oil), annual energy cost is sensitive to electricity price. Assuming £50/MWh renewable power, hydrogen ~\$3–4/kg. At 100k tpa, fuel cost ~£300–400M/year (vs current oil spend ~£600M at £600/tonne). Costs could be higher if grid power is used (instead of dedicated renewables) – sensitivity ±30%.
- **Personnel & Training:** Continue and expand H₂ training programs at Britannia Royal Naval College and Dstl; create new “Fuel Cell Engineer” specialisation. By 2035, all new engineering officers receive hydrogen certification (mirroring 1910s training for oil engineers). Additional key investment: £1–5M/year in personnel training and safety drills.
- **Contingency Plans:** Maintain a small oil-fuel reserve (e.g. continue using marine diesel in emergency generators). Develop bi-fuel engines as a hedge (able to run on H₂ or diesel). Plan for multi-year delays (e.g. if fuel cells underperform, revert some ships to advanced biofuels).

Implementation Timeline (Mermaid Gantt):



5. Operational Impact Assessment

Range and Endurance: Due to hydrogen’s low volumetric density, hydrogen-fuelled vessels will have shorter range *unless* outfitted with much larger tanks. For example, for the same internal fuel volume, a liquid-H₂ fuel-cell ship would carry only ~25–35% of the energy of an oil-fuelled ship (36 MJ/L for diesel vs ~8.5 MJ/L for LH₂, even after accounting for fuel-cell efficiency) [60†L107-L114] [50†L111-L119]. Practically, a destroyer might have its range cut by half or more, or need refuelling twice as often.

Conversely, hydrogen fuel cells and hybrids are quieter and thermally stealthier (no hot exhaust plumes), which could benefit ASW or littoral ops.

Sortie Rates and Flexibility: Refueling with hydrogen is slower with current tech; compressed-H₂ filling requires minutes per kilogram, and LH₂ bunkering demands cryohandling steps. Early hydrogen ships would likely need overnight or longer port fuelling. This may slightly reduce sortie frequency. However, hydrogen generators can be highly reliable: the MOD's H₂ trial generators ran "always-on" charging with zero grid downtime [66†L122-L130] . Similarly, fuel-cell systems have fewer moving parts than reciprocating engines, potentially improving mean time between failures. Overall maintenance would shift from mechanical (fuel pumps, oil change) to electrical/chemical (cell stack upkeep, compressor service).

Safety and Incidents: No major accidents involving shipboard hydrogen propulsion are on record, but the risk profile differs from oil. Hydrogen leaks are hard to detect but can ignite explosively. Safety will rely on robust detection and ventilation. By contrast, oil fires are easier to see but harder to extinguish. Navy firefighting procedures would need revision (the MoD trial reported burning hydrogen would *only* produce water, easing fire control). Personnel risk shifts from coal-dust explosions and oil-blazes (1913-era) to high-pressure gas leaks and cryogenic burns. Close coordination with regulatory bodies (e.g. Lloyd's Register, MCA) will be essential to set safety standards.

Environmental Benefits: Hydrogen fuel (if green) yields virtually zero CO₂ and negligible pollutants. This reduces the Navy's carbon footprint to near-zero for operations, aligning with UK climate policy and potentially easing political/public pressure. Zero SO_x/NO_x emissions will also reduce shipboard and portside pollution (unlike 1914, when oil burning already eliminated coal smoke but still emitted SO_x). The MCA notes hydrogen is "zero-carbon" fuel [57†L96-L104] . Any nitrogen oxides from combustion would be modest compared to diesel. Overall, the environmental impact is strongly positive: less greenhouse gas, no soot, no sulfur emissions.

6. Risks, Mitigations & Recommendations

- **Supply Vulnerability:** Relying on hydrogen imports (or single-site production) could be a strategic risk. *Mitigation:* diversify suppliers (domestic electrolysis, renewables-powered imports, ammonia carriers), and maintain emergency stockpiles (drawing a lesson from coal reserves). Establish cross-allied fuel sharing (e.g. UK-US NATO hydrogen stock agreements).
- **Technology Maturity:** Fuel-cell and hydrogen storage tech for large ships is still developing. *Mitigation:* invest in R&D and incremental testing now, in collaboration with industry (as with the 1912 Commission's trials). Fund a UK naval hydrogen lab to accelerate learning curves. Keep conventional engines as backups through the transition.
- **Industrial and Cost Risks:** Upfront costs are high and budget uncertain. *Mitigation:* phase spending over a decade, use public-private partnerships, and leverage civilian marine hydrogen projects (e.g. adapt commercial ferry developments). Perform regular cost reviews. Sensitivity analysis should consider electricity price swings and CAPEX inflation.
- **Safety and Regulatory:** Inadequate rules could lead to accidents or delays. *Mitigation:* the RN should proactively engage IMO/MCA to shape hydrogen codes (the 1912 US Commission report recommended codifying fuel-oil use, and today's equivalent is fast-tracking the IGF Code for hydrogen). Develop strict RN instructions for handling hydrogen (building on MOD's EV charging trial emphasis on training [66†L150-L154]).
- **Training and Doctrine:** There is a risk of skill gaps. *Mitigation:* rollout comprehensive training early (simulate hydrogen emergencies in exercises). The MoD's trial suggests training is key to

“decarbonise power use” in the armed forces 【66†L150-L154】 . Allocate permanent hydrogen training billets in engineering branches.

- **Operational Risk:** A partial transition could leave mixed fleets. *Mitigation:* ensure interoperability between old and new (e.g. fuel cross-feed lines). As in 1914, keep coal (or here, diesel) engines in reserve until confidence is absolute. Consider hybrid vessels that can burn either fuel type as needed.

Policy Recommendations: The MOD should commit to a clear hydrogen roadmap (with associated budget) and avoid “stop-go” procurement. Provide incentives to the UK supply chain (shipyards, energy firms) for hydrogen projects, similar to how the 1914 Admiralty encouraged oil exploration and storage 【16†L42-L50】 . Engage Parliament early to build consensus (as Churchill did for the AP Oil purchase). Coordinate with the Department for Transport and Energy on national hydrogen infrastructure so Navy needs are met. Finally, establish “green hydrogen supply credits” or similar accounting to ensure claimed emissions reductions are real (avoiding “blue-wash”).

Table – Fuel Comparison: The table below compares coal, conventional fuel oil and hydrogen along key attributes (energy density, infrastructure, emissions, etc). This highlights the challenges of transitioning from a heavy liquid fuel to a very light one:

Attribute	Coal	Oil (Marine Diesel)	Hydrogen (H ₂)
Phase/Tank Type	Solid (sack/bin)	Liquid (tank)	Gas/Liquid (high-pressure or cryo tank)
Energy Density	~24 MJ/kg (19 MJ/L)	~45 MJ/kg (36 MJ/L)	120 MJ/kg but ~8.5 MJ/L (LH ₂) or ~5 MJ/L (gaseous)
Storage Weight	Heavy (coal)	Moderate (fuel oil ~0.85 kg/L)	Light by mass, but tank insulation heavy
Refueling Infrastructure	Extensive 19th-c global coaling bases	Modern bunkering ports worldwide	Almost nonexistent now; new LH ₂ /ammonia terminals needed
Operational Handling	Manual shoveling (labor-intensive)	Pumped by engine room staff (common practice)	Compressed gas or cryo handling, new procedures, high tech
Logistics	Inland mines & shipping	Global oil markets (Russia, Middle East, USA)	Emerging renewables-based supply (UK wind, imports, etc)
Emissions (CO₂)	High (fossil)	High (fossil)	Zero-carbon at point of use (if green H ₂); potential CO ₂ if blue production
Other Emissions	SO _x , NO _x , ash	NO _x (diesel)	Essentially none (only water, trace NO _x from combustion)
Safety Hazards	Fire/explosion (coal dust)	Fire (oil fires)	Flammability (H ₂ gas explosion risk), cryogenic burns; (if ammonia used: toxicity)

Attribute	Coal	Oil (Marine Diesel)	Hydrogen (H ₂)
Current Fleet Use	Phased out by WWII (except legacy)	Primary naval fuel (to present)	Prototype stage; no current large-ship use
Historical Shift (1913)	Obsolete soon after transition	New standard by WWI	–
Prospects (2027)	Irrelevant	Transitional fuel	Potential future fuel

This table (based on technical data and the sources above) underscores that while hydrogen eliminates carbon emissions, it introduces major changes in storage and infrastructure.

Sources

We draw on official and scholarly sources. Admiralty/Hansard records from 1912–14 provide firsthand detail of the coal-to-oil decision (e.g. Churchill’s Commons speech on Anglo-Persian Oil [【12†L19-L27】](#) , Fisher’s report summary [【16†L42-L50】](#) , MPs’ debate on risks [【14†L4001-L4004】](#)). The UK Maritime & Coastguard Agency’s 2025 guidance outlines hydrogen’s fuel properties and regulatory status [【57†L96-L104】](#) . Recent industry and defence publications inform the hydrogen analysis: for example, the UK MOD’s GeoPura trial report (Sept 2024) highlights hydrogen’s feasibility and MoD’s learning curve [【66†L114-L120】](#) [【66†L146-L154】](#) . Technical journals and white papers (e.g. *Fuel*, *International Journal of Hydrogen Energy*) were surveyed for vessel performance metrics; where peer-reviewed citations are unavailable we rely on credible industry sources (e.g. Fuel Cell Association summaries [【60†L81-L88】](#) [【60†L107-L114】](#)). All cited passages are linked above in the [【...†L...】](#) format for verification.